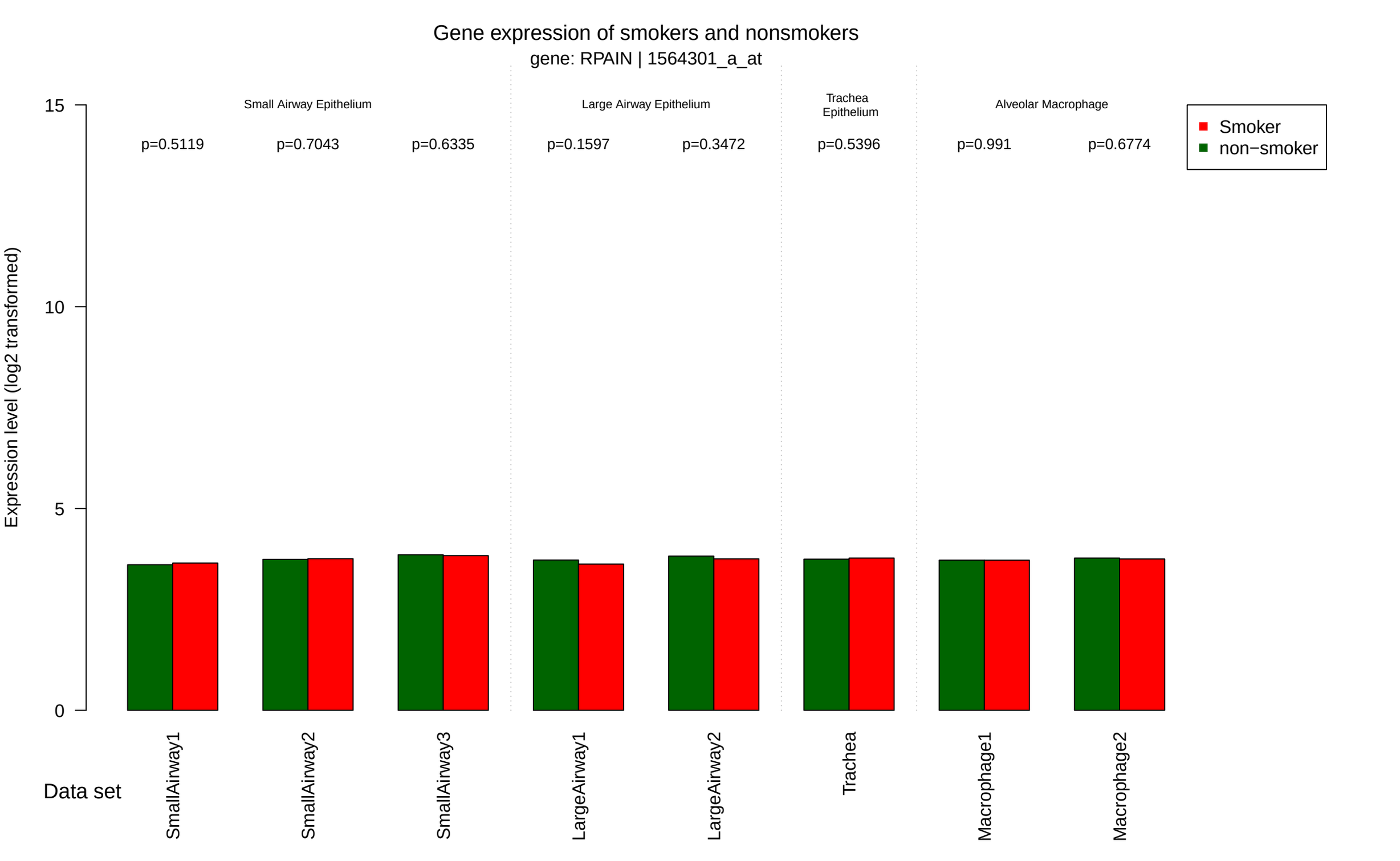
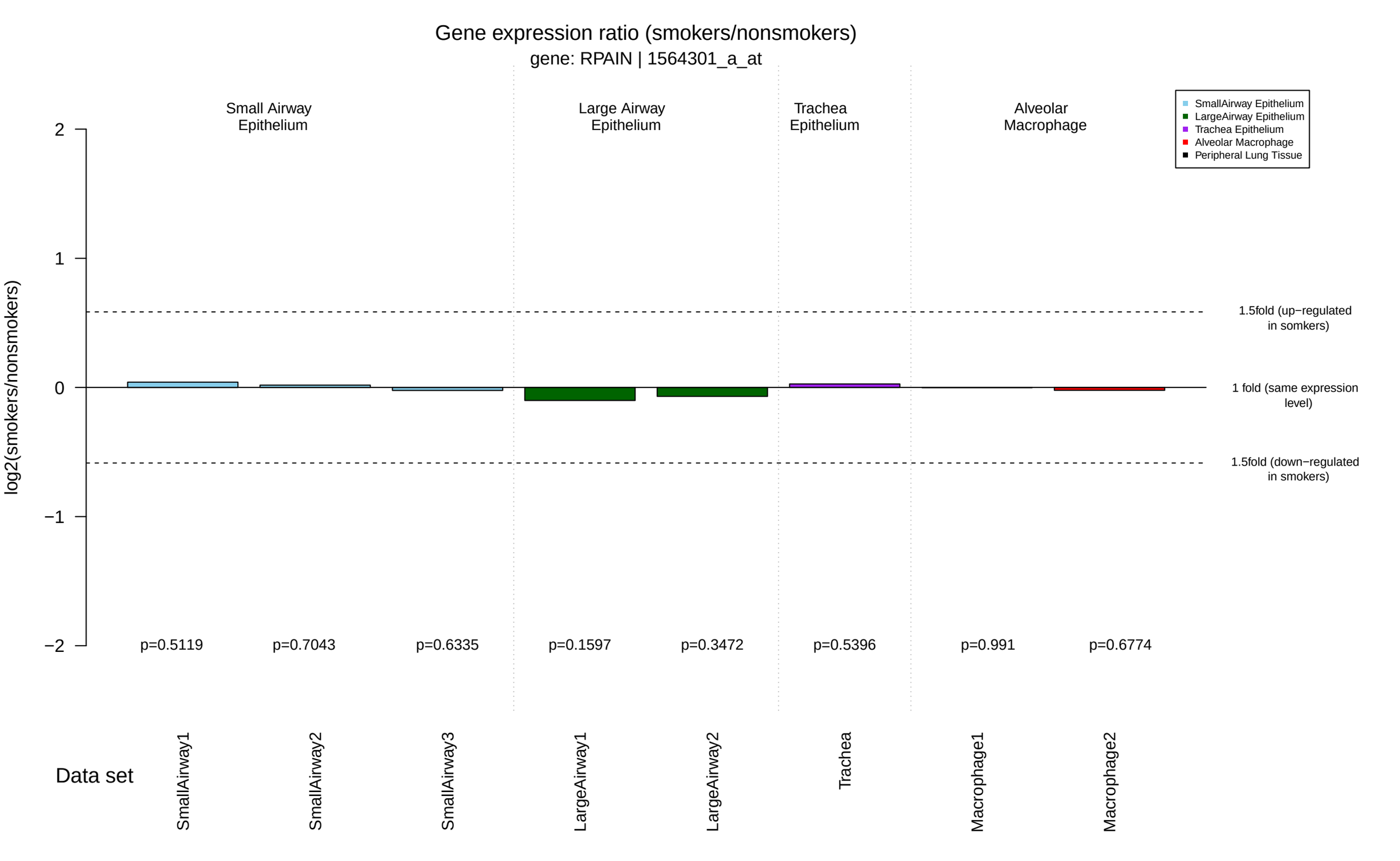
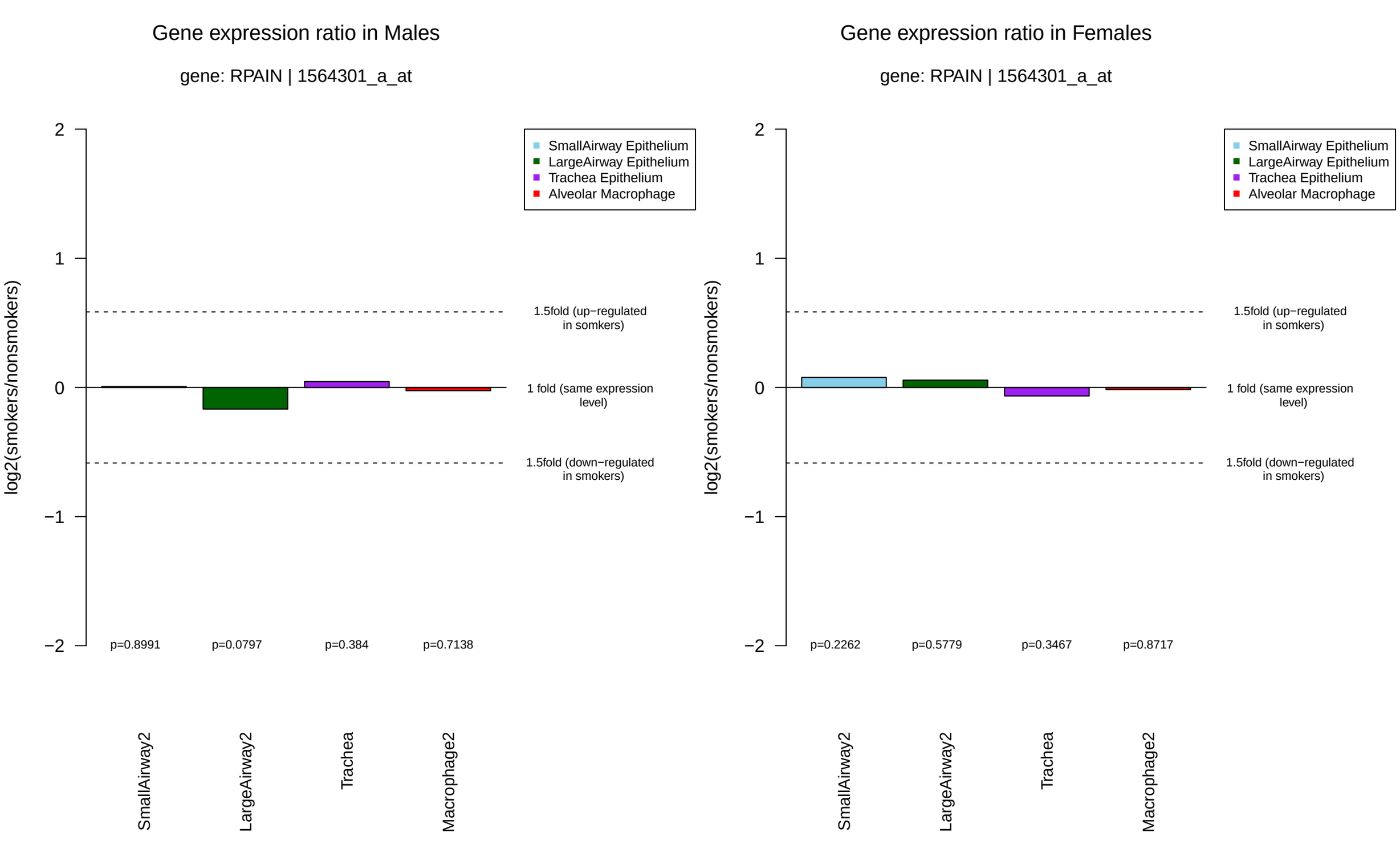
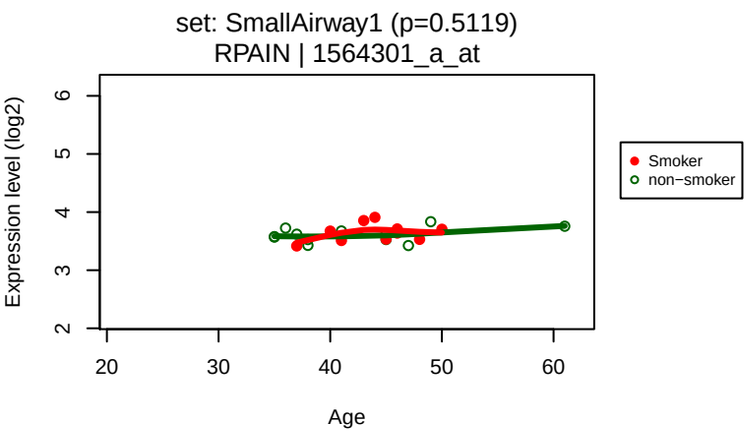
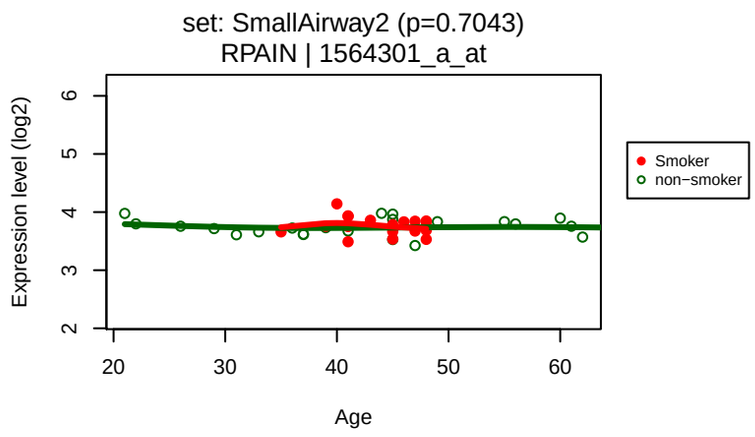
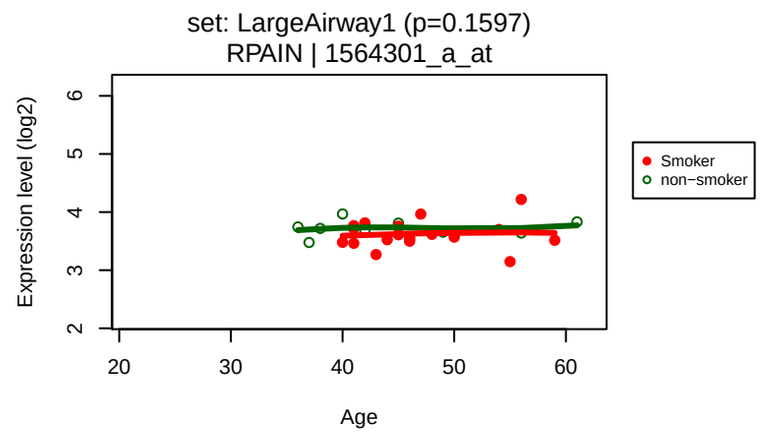

Gene expression of smokers and nonsmokers
gene: RPAIN | 1564301_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5119
p=0.7043
p=0.6335
p=0.1597
p=0.3472
p=0.5396
p=0.991
p=0.6774
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

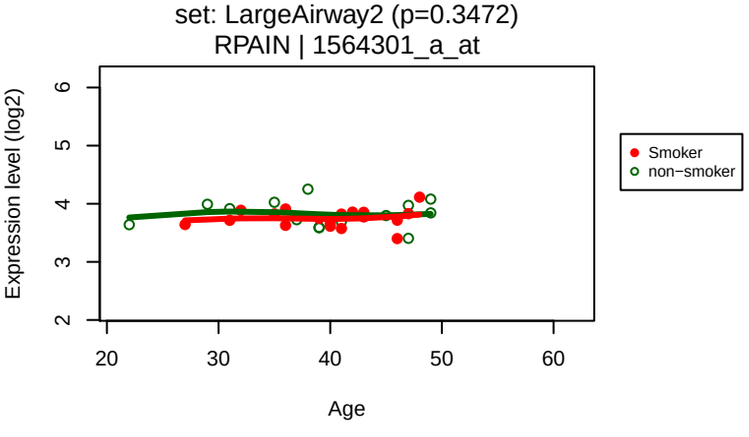

Gene expression ratio (smokers/nonsmokers)
gene: RPAIN | 1564301_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5119
p=0.7043
p=0.6335
p=0.1597
p=0.3472
p=0.5396
p=0.991
p=0.6774
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

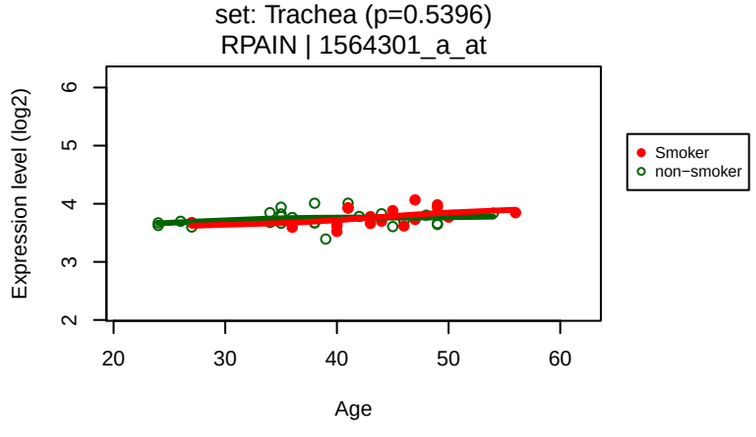

Gene expression ratio in Males
gene: RPAIN | 1564301_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8991
p=0.0797
p=0.384
p=0.7138
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPAIN | 1564301_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2262
p=0.5779
p=0.3467
p=0.8717
SmallAirway2
LargeAirway2
Trachea
Macrophage2

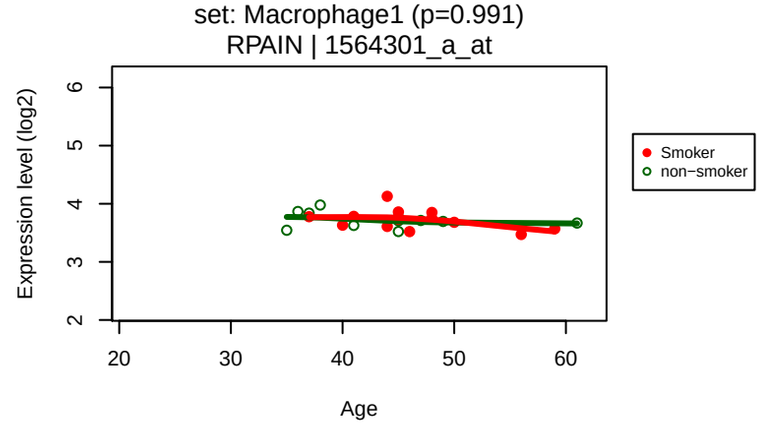

set: SmallAirway1 (p=0.5119)
RPAIN | 1564301_a_at
Expression level (log2)
Age
Smoker
non−smoker

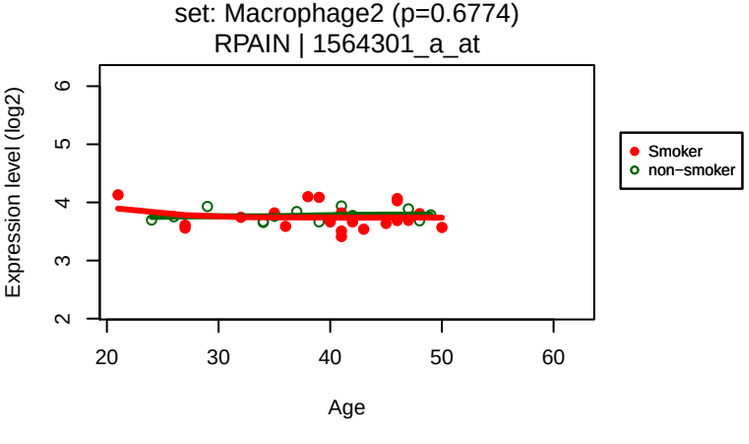

set: SmallAirway2 (p=0.7043)
RPAIN | 1564301_a_at
Expression level (log2)
Age
Smoker
non−smoker

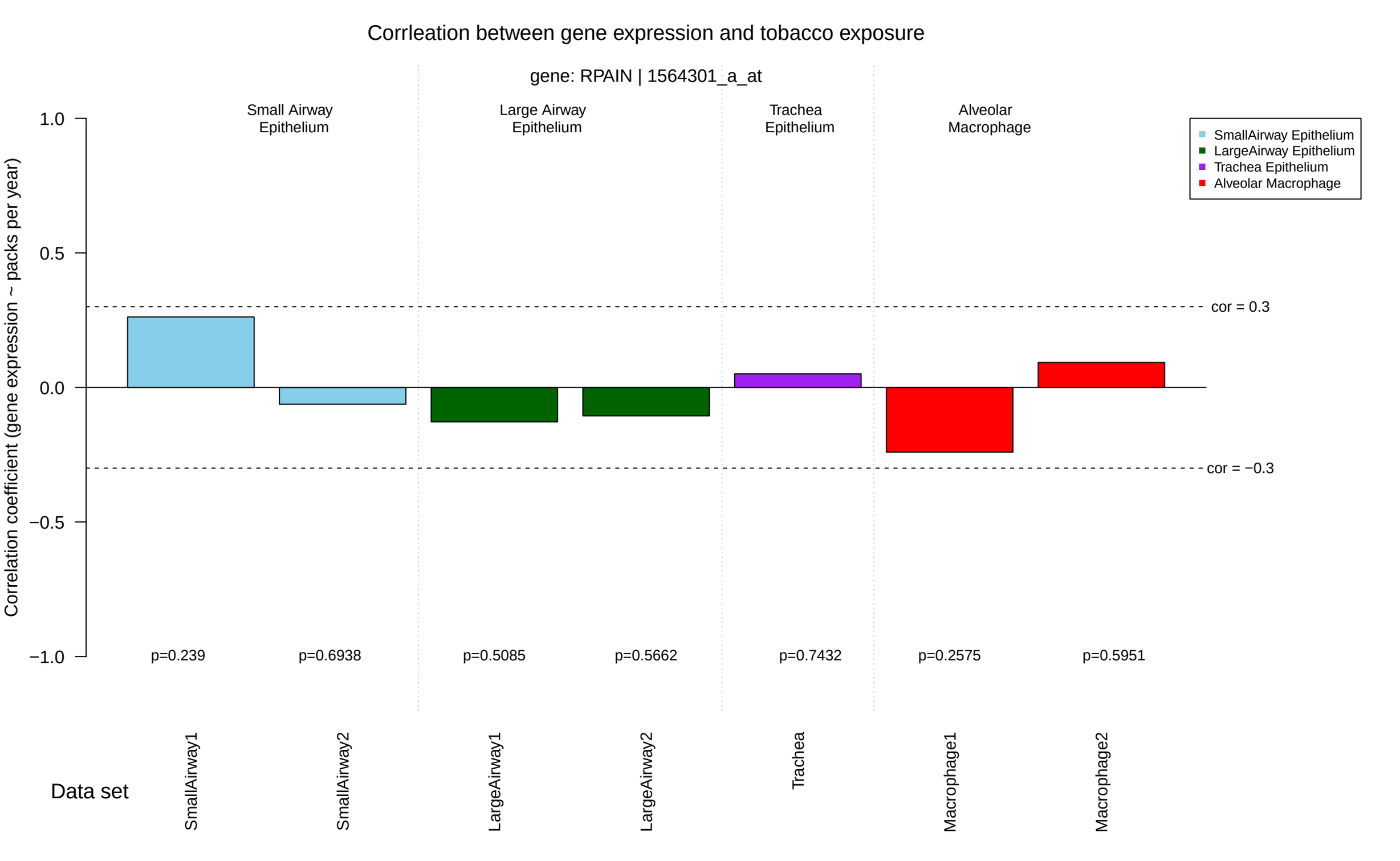

set: LargeAirway1 (p=0.1597)
RPAIN | 1564301_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3472)
RPAIN | 1564301_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5396)
RPAIN | 1564301_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.991)
RPAIN | 1564301_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6774)
RPAIN | 1564301_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPAIN | 1564301_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.239
p=0.6938
p=0.5085
p=0.5662
p=0.7432
p=0.2575
p=0.5951
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2